

How the six laboratories work

ORIGINAL ILLUSTRATIVE PLAY

These are original engine-assisted study continuations from this course's positions. They are not historical master games and no invented players or tournament results are assigned. The continuations reach roughly move twenty and stop for further practice.

THINK BEFORE REVEALING

At each checkpoint, name the opponent's threat, compare two candidates and calculate a forcing reply. Use a board to replay the sequence. The engine comparison supplies concrete evidence, while the opening explanation supplies a plan.

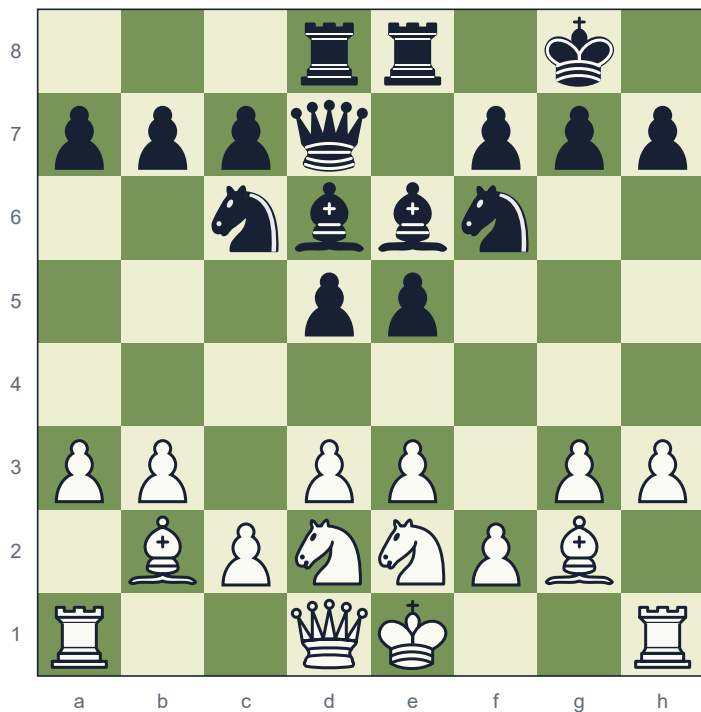
MORE THAN ONE GOOD MOVE

Close engine scores do not establish a unique answer. The finite search can prefer a different move when run longer. Accept another candidate if you can support it with a sound variation and a coherent explanation.

PRACTICE FROM BOTH SIDES

Play each starting position twice, switching colors. Record the first decision you could not explain and compare it with the branch lesson after the game.

HW-LAB-01 / The complete formation as a study position



Laboratory start - White to move

OPENING BACKGROUND

White reaches the familiar two-rank formation while Black occupies the center. This quiet illustrative route is not a claim that all ten setup moves are best in every position.

READ THE CURRENT POSITION

White king e1; central-file pawns c2, f2, d3, e3. Black king g8; central-file pawns d5, e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

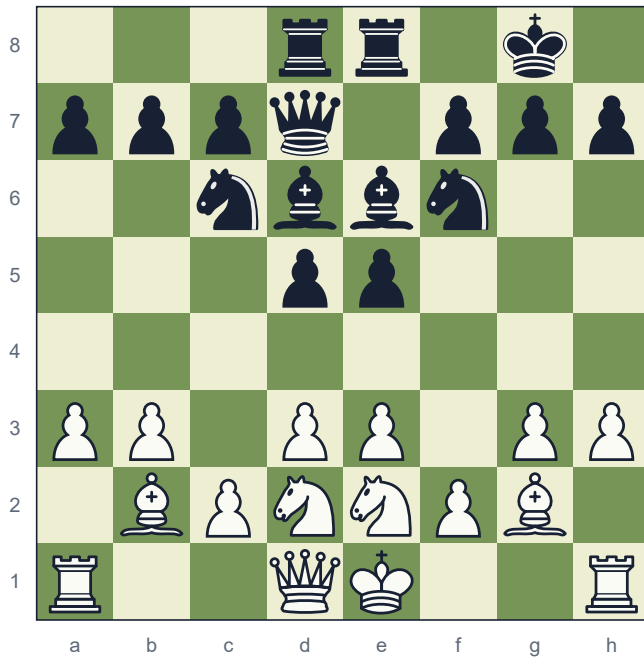
Black has more space and can prepare ...e4 or ...d4. White's extra tempo has not prevented a central bind. Decide which parts still apply after the moves already played.

BEFORE PLAYING

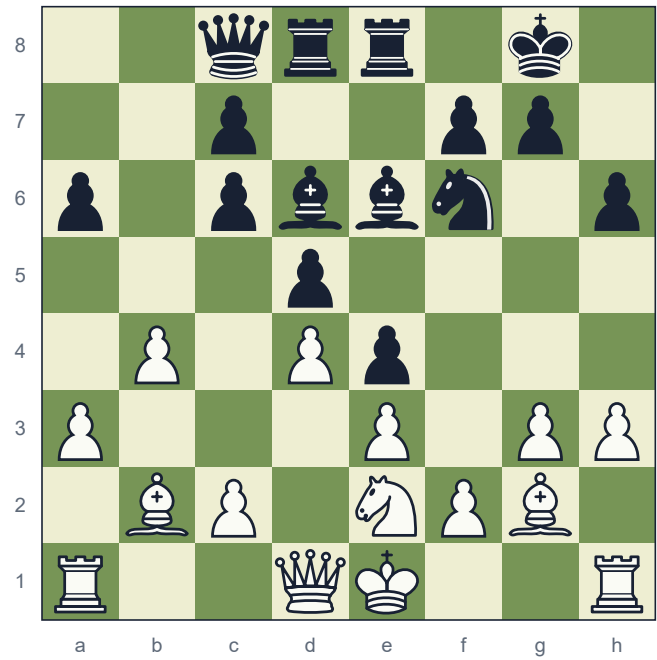
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-01 / play and explain 1

11. Nf3 h6 12. b4 a6 13. d4 e4 14. Ne5 Qc8 15. Nxc6 bxc6



Before this segment - White to move



After this segment - White to move

11. Nf3

Develops the king knight toward e5 and d4. In the King's Gambit it also controls h4, making an immediate queen check there unattractive.

13. D4

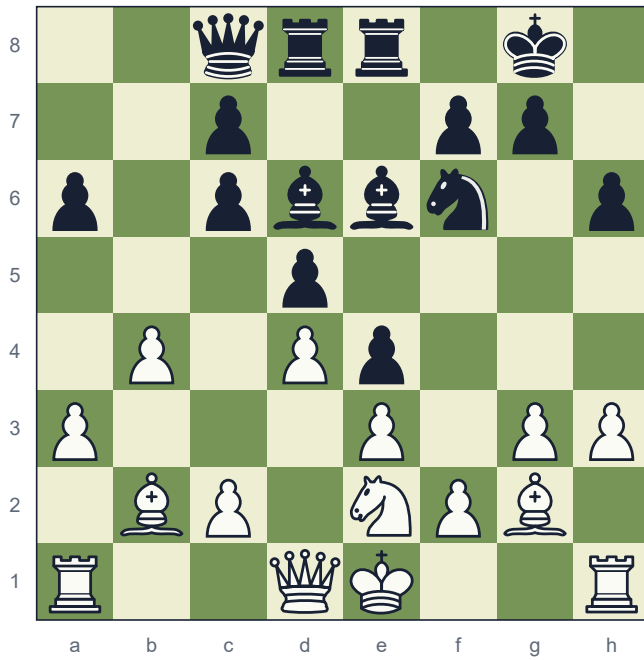
Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

REASSESS

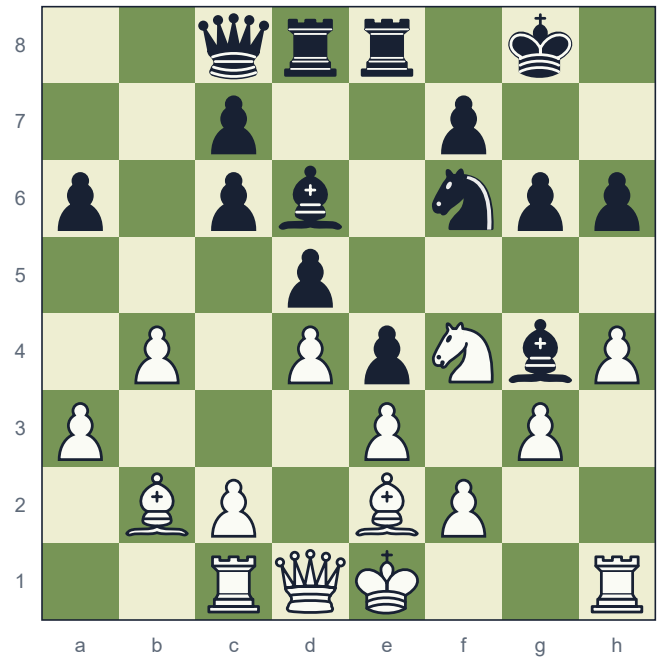
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-01 / play and explain 2

16. Nf4 Bf5 17. Bf1 Nh7 18. h4 Nf6 19. Be2 g6 20. Rc1 Bg4



Before this segment - White to move



After this segment - White to move

16. Nf4

Places the knight on f4, attacking bishop on e6. A legal capture may still be unsound if the target is defended.

18. H4

Creates a wing or central pawn contact and changes king shelter. Calculate captures and the opponent's central counterplay before extending the advance.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-01 / decision 1



White to move

COMPARE CANDIDATES

Nf3 and g4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king e1; central-file pawns c2, f2, d3, e3. Black king g8; central-file pawns d5, e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NF3

11. Nf3 h6 12. d4 Bf5 13. dxe5 Nxe5 14. Nxe5 Bxe5 Search estimate: -0.95 from White's perspective.

CANDIDATE G4

11. g4 h6 12. Ng3 Ne7 13. Nf3 d4 14. g5 Nfd5 Search estimate: -1.01 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Develops the king knight toward e5 and d4. In the King's Gambit it also controls h4, making an immediate queen check there unattractive.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-01 / decision 2



White to move

COMPARE CANDIDATES

Nxc6 and Nf4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king e1; central-file pawns c2, f2, e3, d4. Black king g8; central-file pawns e4, d5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NXC6

15. Nxc6 bxc6 16. Nf4 Bf5 17. Qe2 Qb7 18. O-O Qc8 Search estimate: -1.15 from White's perspective.

CANDIDATE NF4

15. Nf4 Bf5 16. Nxc6 bxc6 17. Qe2 Qb7 18. Bc3 Nh7 Search estimate: -1.16 from White's perspective.

EXPLAIN THE PREFERRED MOVE

The knight captures a knight on c6. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-01 / decision 3



White to move

COMPARE CANDIDATES

Be2 and c4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king e1; central-file pawns c2, f2, e3, d4. Black king g8; central-file pawns e4, d5, c6, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BE2

19. Be2 g6 Search estimate: -1.07 from White's perspective.

CANDIDATE C4

19. c4 Bg4 20. Be2 dxc4 21. Bxg4 Nxg4 22. Rc1 a5 Search estimate: -1.37 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Places the bishop on e2. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-02 / Black grabs space with ...c5



Laboratory start - White to move

OPENING BACKGROUND

Black establishes c5-d5-e5. White must challenge the center before completing every wing pawn move.

READ THE CURRENT POSITION

White king g1; central-file pawns c2, d3, e3, f4. Black king e8; central-file pawns e4, c5, d5, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

...d4 can close the center with a large space advantage. If every white piece lacks squares, the setup has become too slow. Decide which parts still apply after the moves already played.

BEFORE PLAYING

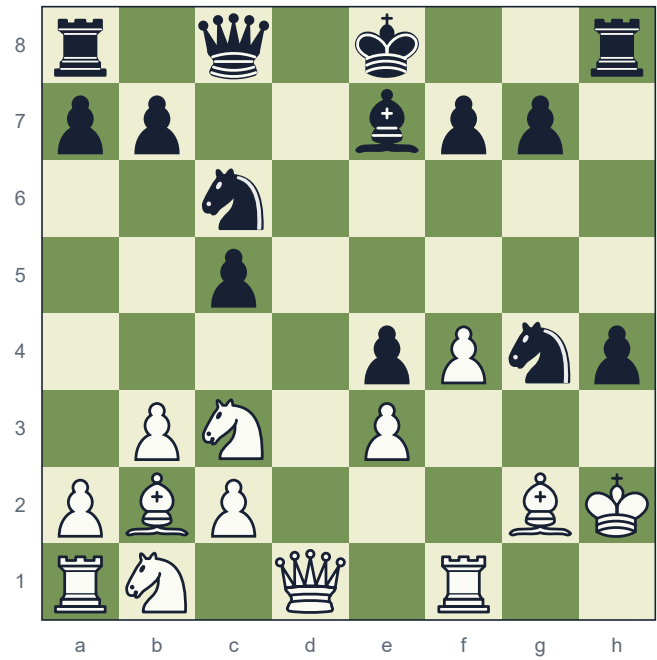
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-02 / play and explain 1

11. dxe4 dxe4 12. Nec3 Qc8 13. Kh2 h4 14. g4 Bxg4 15. hxg4 Nxg4+



Before this segment - White to move



After this segment - White to move - in check

11. DXE4

The pawn captures a pawn on e4. Calculate the best reply before assuming a recapture, then recount material and opened lines.

13. KH2

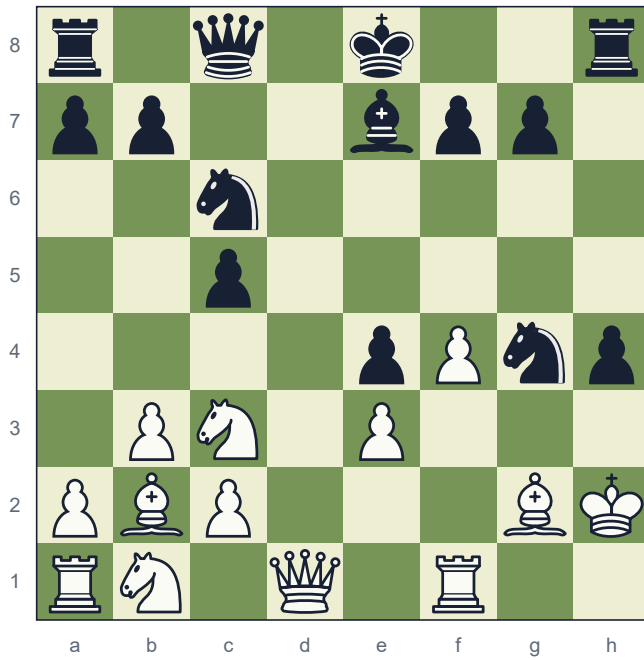
Places the king on h2. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

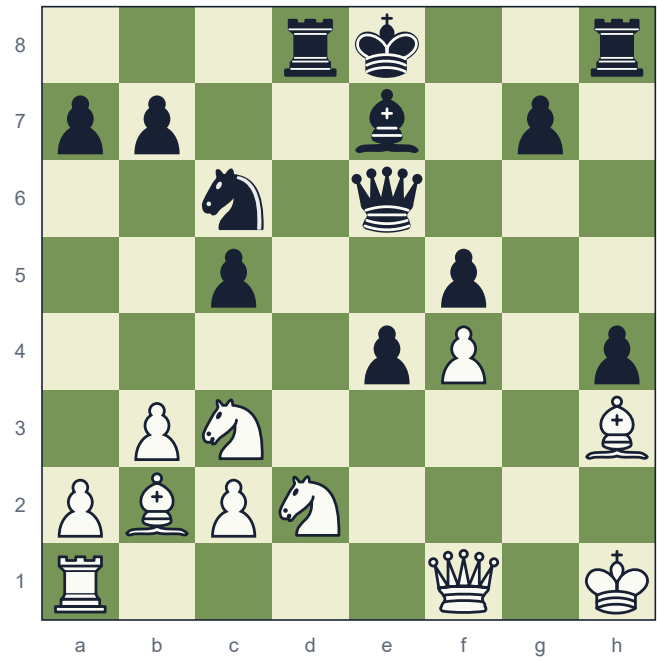
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-02 / play and explain 2

16. Kh1 Nxe3 17. Qe2 Nxf1 18. Qxf1 f5 19. Bh3 Qe6 20. Nd2 Rd8



Before this segment - White to move - in check



After this segment - White to move

16. KH1

Places the king on h1. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

18. QXF1

The queen captures a knight on f1. Calculate the best reply before assuming a recapture, then recount material and opened lines.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-02 / decision 1



COMPARE CANDIDATES

dxe4 and Bxf6. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, d3, e3, f4. Black king e8; central-file pawns e4, c5, d5, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE DXE4

11. dxe4 dxe4 12. Nec3 Rd8 13. Nd2 Qc8 14. Qe2 Bxh3 Search estimate: -0.60 from White's perspective.

CANDIDATE BXF6

11. Bxf6 Bxf6 Search estimate: -1.37 from White's perspective.

EXPLAIN THE PREFERRED MOVE

The pawn captures a pawn on e4. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-02 / decision 2



White to move

COMPARE CANDIDATES

hxc4 and Qd2. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king h2; central-file pawns c2, e3, f4. Black king e8; central-file pawns e4, c5, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE HXG4

15. hxc4 Nxc4+ Search estimate: -0.39 from White's perspective.

CANDIDATE QD2

15. Qd2 Bf5 16. Nb5 O-O 17. Nd6 Bxd6 18. Bxf6 Be7 Search estimate: -1.24 from White's perspective.

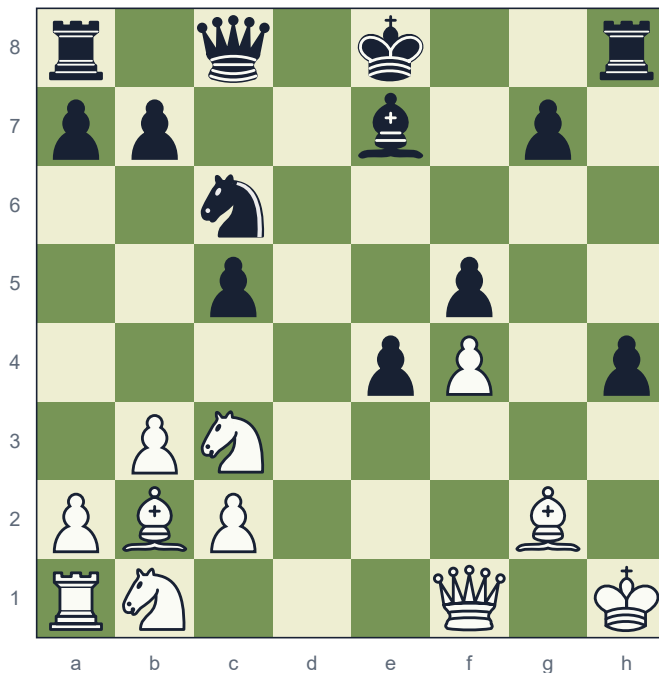
EXPLAIN THE PREFERRED MOVE

The pawn captures a bishop on g4. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-02 / decision 3



COMPARE CANDIDATES

Bh3 and Qf2. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king h1; central-file pawns c2, f4. Black king e8; central-file pawns e4, c5, f5. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BH3

19. Bh3 Search estimate: -0.24 from White's perspective.

CANDIDATE QF2

19. Qf2 Nd4 20. Nd5 h3 21. Bf1 Qc6 22. Bxd4 Qxd5 Search estimate: -2.00 from White's perspective.

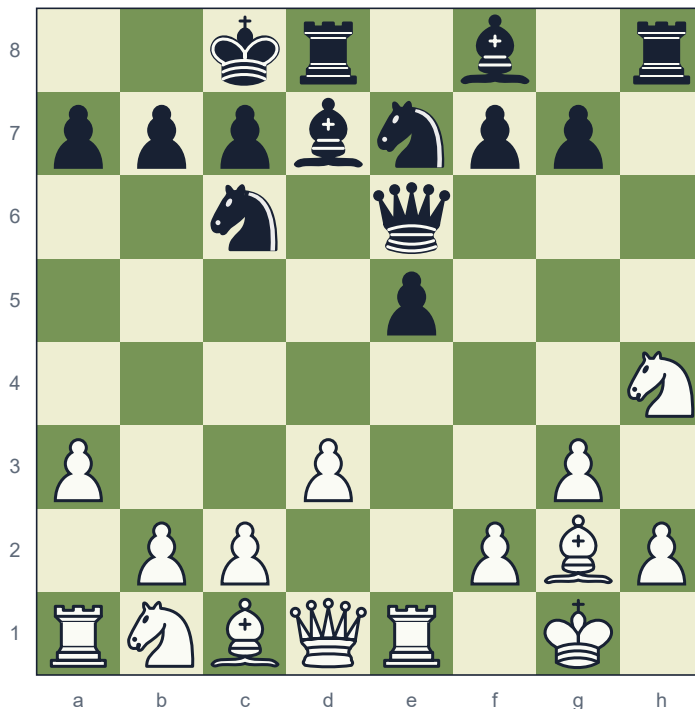
EXPLAIN THE PREFERRED MOVE

Places the bishop on h3. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-03 / An immediate ...h5



Laboratory start - White to move

OPENING BACKGROUND

Black attacks the g3 pawn before building a center. Nf3 develops and can capture on h4 when Black advances without support.

READ THE CURRENT POSITION

White king g1; central-file pawns c2, f2, d3. Black king c8; central-file pawns e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

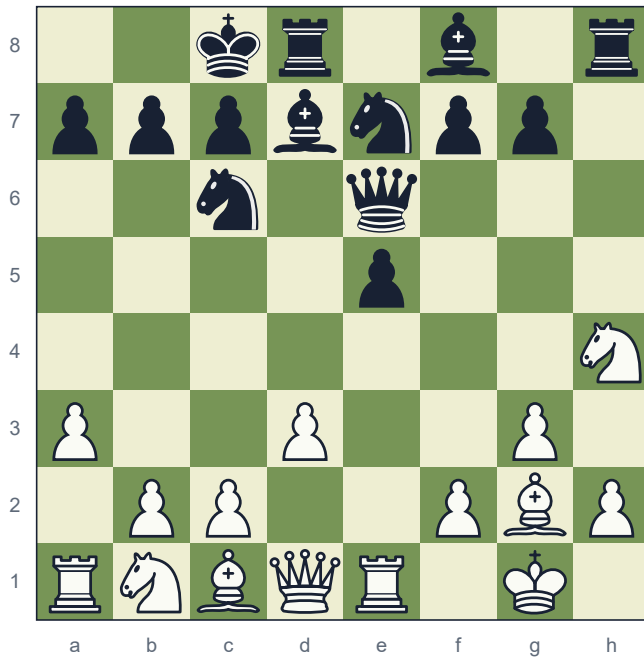
Black may strike in the center while the knight sits on the rim. Do not keep chasing pawns instead of developing. Decide which parts still apply after the moves already played.

BEFORE PLAYING

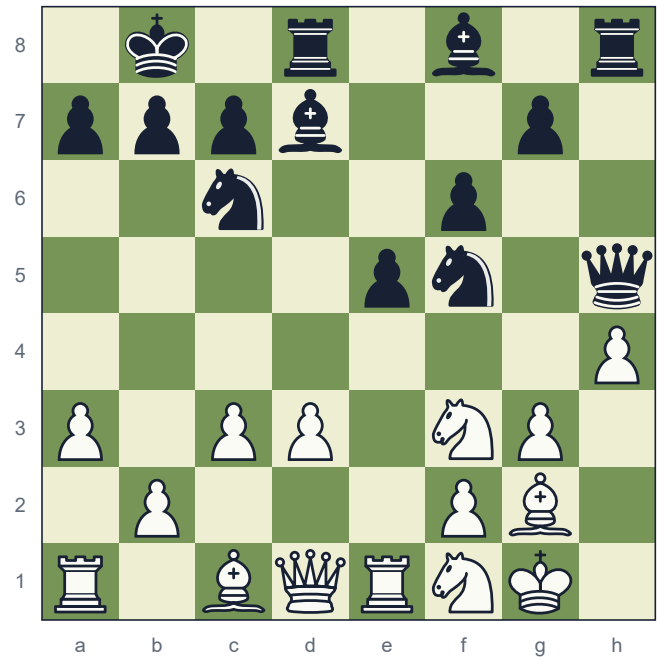
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-03 / play and explain 1

11. Nf3 Qf5 12. h4 Qh5 13. Nbd2 f6 14. Nf1 Nf5 15. c3 Kb8



Before this segment - White to move



After this segment - White to move

11. NF3

Develops the king knight toward e5 and d4. In the King's Gambit it also controls h4, making an immediate queen check there unattractive.

13. NBD2

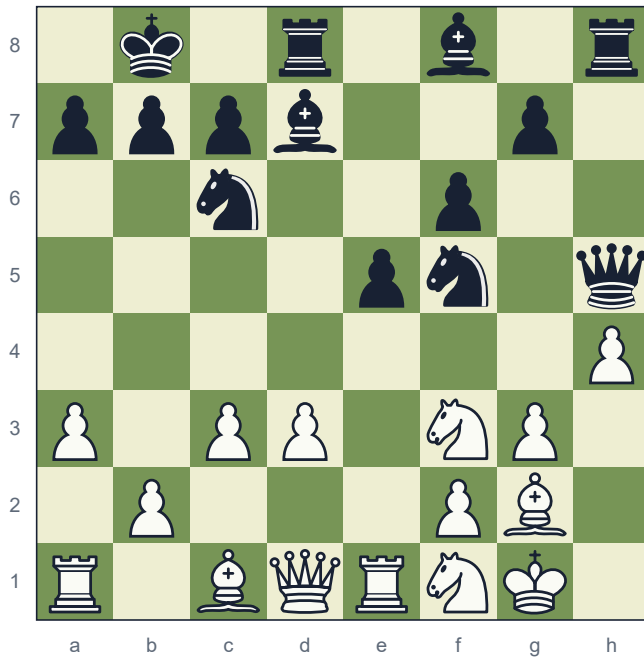
Develops the queen knight while leaving the c-pawn free. The knight supports e4 and may seek c4 or a route through f1.

REASSESS

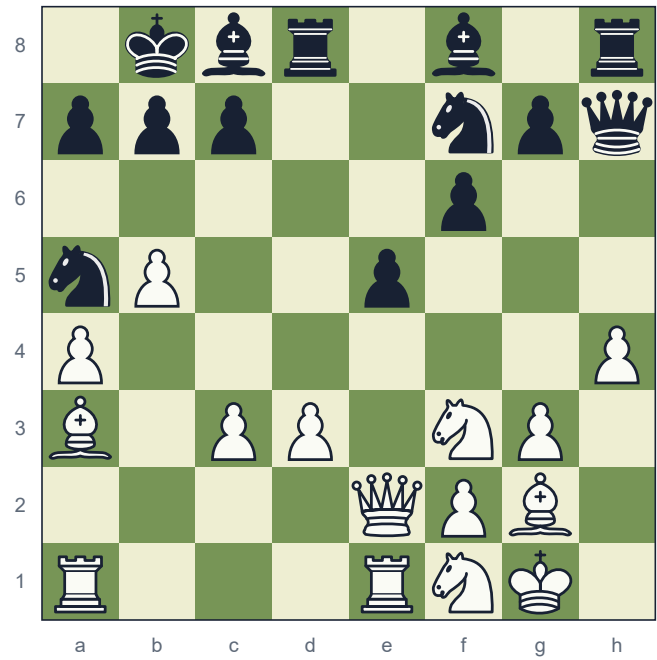
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-03 / play and explain 2

16. b4 Nd6 17. a4 Bc8 18. Ba3 Qh7 19. b5 Na5 20. Qe2 Nf7



Before this segment - White to move



After this segment - White to move

16. B4

Places the pawn on b4. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

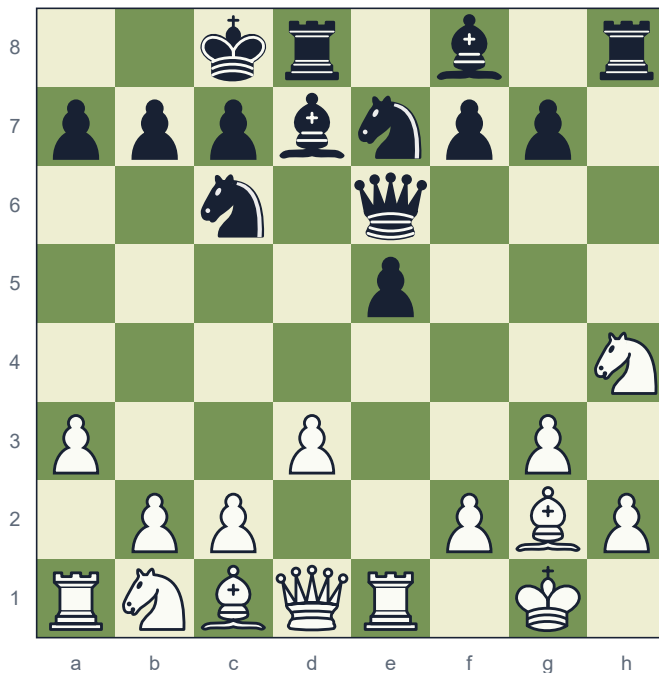
18. BA3

Places the bishop on a3. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-03 / decision 1



White to move

COMPARE CANDIDATES

Nf3 and b4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d3. Black king c8; central-file pawns e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NF3

11. Nf3 Qf5 Search estimate: +0.68 from White's perspective.

CANDIDATE B4

11. b4 Rxh4 12. gxh4 Nf5 13. Re4 Nd6 14. Re1 Search estimate: +0.00 from White's perspective.

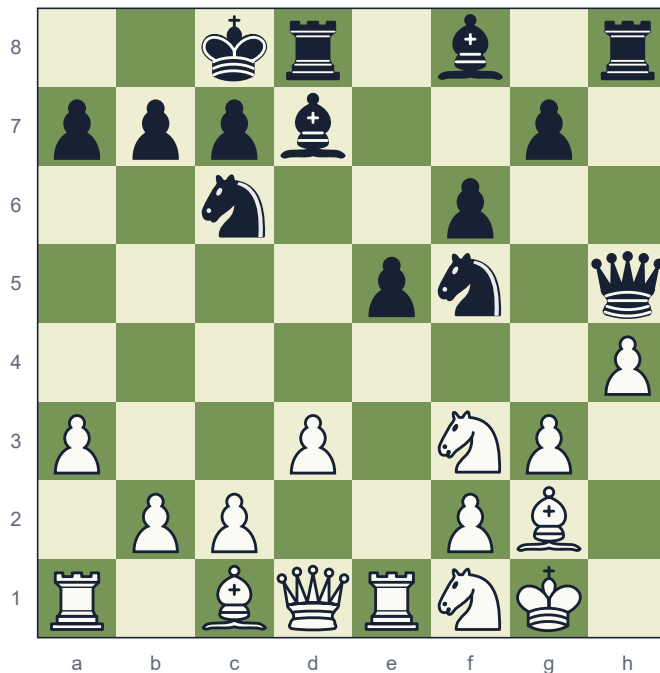
EXPLAIN THE PREFERRED MOVE

Develops the king knight toward e5 and d4. In the King's Gambit it also controls h4, making an immediate queen check there unattractive.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-03 / decision 2



COMPARE CANDIDATES

c3 and b4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d3. Black king c8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE C3

15. c3 Kb8 16. b4 Nd6 17. c4 Bf5 18. Ne3 Bc8 Search estimate: +1.09 from White's perspective.

CANDIDATE B4

15. b4 Ncd4 16. Be3 Nxf3+ 17. Qxf3 Qxf3 18. Bxf3 g6 Search estimate: +0.91 from White's perspective.

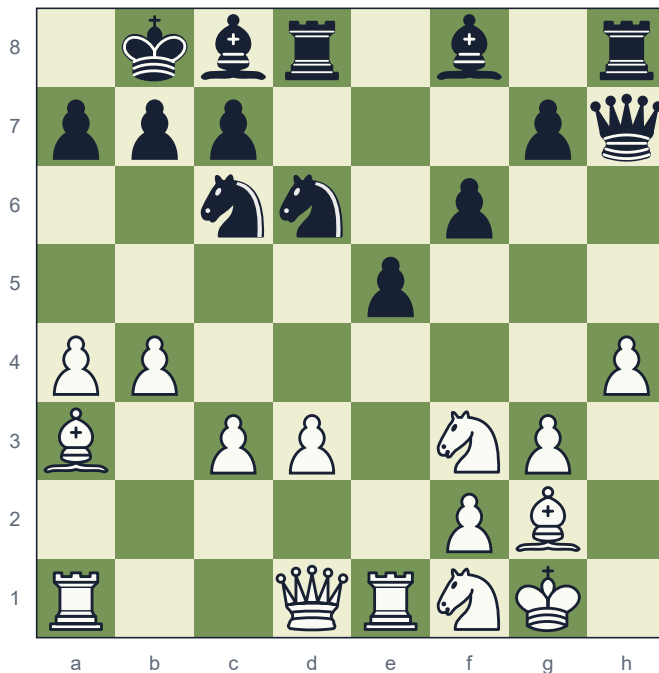
EXPLAIN THE PREFERRED MOVE

Places the pawn on c3. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-03 / decision 3



White to move

COMPARE CANDIDATES

b5 and Qe2. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c3, d3. Black king b8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE B5

19. b5 Search estimate: +1.06 from White's perspective.

CANDIDATE QE2

19. Qe2 Nf7 20. b5 Na5 21. Bxf8 Rdx8 22. d4 g5 Search estimate: +0.99 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Places the pawn on b5, attacking knight on c6. A legal capture may still be unsound if the target is defended.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-04 / An early ...Bh3 attempt



Laboratory start - White to move

OPENING BACKGROUND

Black exchanges the g2 bishop and brings the queen toward White's already castled king. White's knight has left g1, so the queen on h3 cannot simply be captured by that knight.

READ THE CURRENT POSITION

White king g1; central-file pawns c2, f2, d3. Black king e8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

If White's knight were still on g1, ...Qxh3 would lose the queen to Nxh3. In this line its different square makes the attack a real concern. Decide which parts still apply after the moves already played.

BEFORE PLAYING

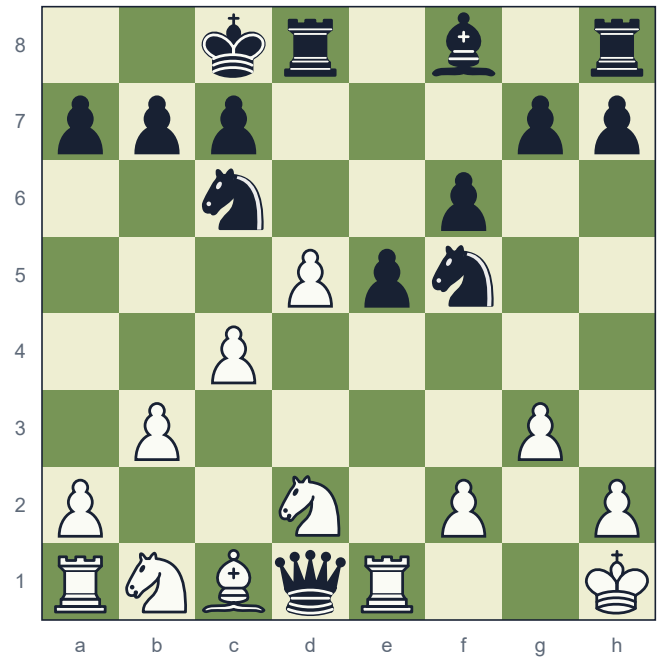
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-04 / play and explain 1

11. d4 O-O-O 12. c4 Nde7 13. d5 Nf5 14. Kh1 Qg4 15. Nfd2 Qxd1



Before this segment - White to move



After this segment - White to move

11. D4

Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

13. D5

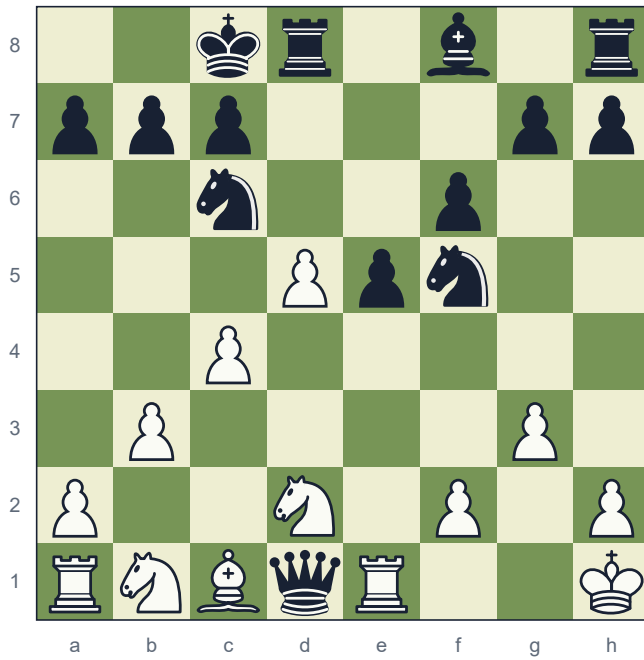
Places the pawn on d5, attacking knight on c6. A legal capture may still be unsound if the target is defended.

REASSESS

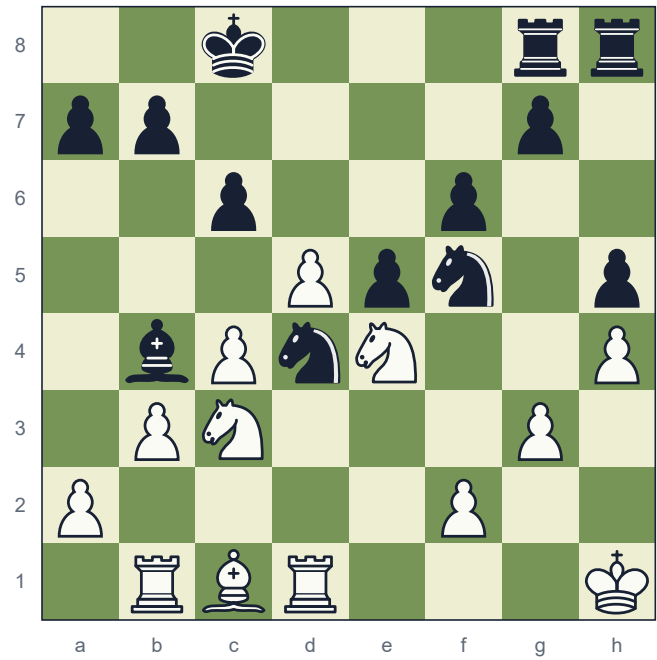
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-04 / play and explain 2

16. Rxd1 Ncd4 17. Nc3 h5 18. Nde4 c6 19. Rb1 Bb4 20. h4 Rdg8



Before this segment - White to move



After this segment - White to move

16. RxD1

The rook captures a queen on d1. Calculate the best reply before assuming a recapture, then recount material and opened lines.

18. NDE4

Places the knight on e4. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-04 / decision 1



COMPARE CANDIDATES

d4 and c4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns c2, f2, d3. Black king e8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE D4

11. d4 Search estimate: -0.66 from White's perspective.

CANDIDATE C4

11. c4 Nde7 12. b4 Nf5 13. b5 Ncd4 14. Nxd4 Nxd4 Search estimate: -1.17 from White's perspective.

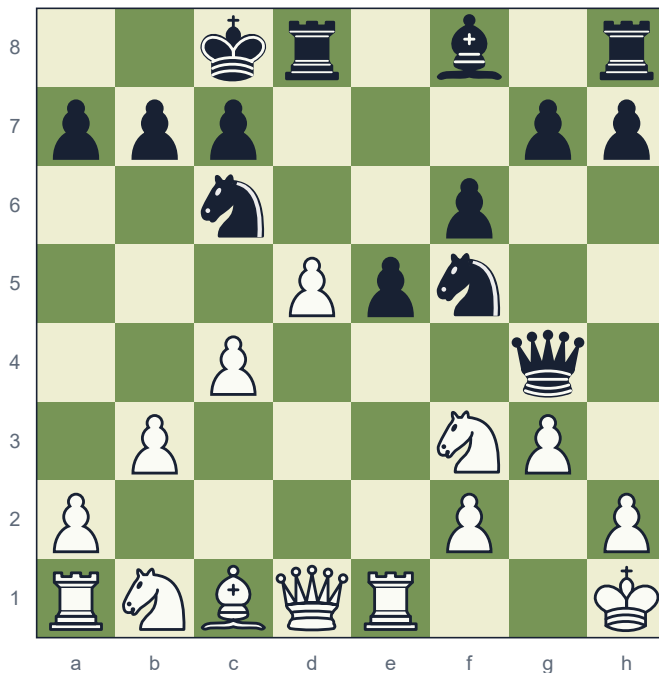
EXPLAIN THE PREFERRED MOVE

Changes central space and pawn contacts. Calculate captures, advances and maintaining tension before choosing the next piece plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-04 / decision 2



White to move

COMPARE CANDIDATES

Nfd2 and Nbd2. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king h1; central-file pawns f2, c4, d5. Black king c8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NFD2

15. Nfd2 Qxd1 16. Rxd1 Ncd4 17. Nc3 Be7 18. Rb1 h5 Search estimate: -0.38 from White's perspective.

CANDIDATE NBD2

15. Nbd2 Ncd4 Search estimate: -0.44 from White's perspective.

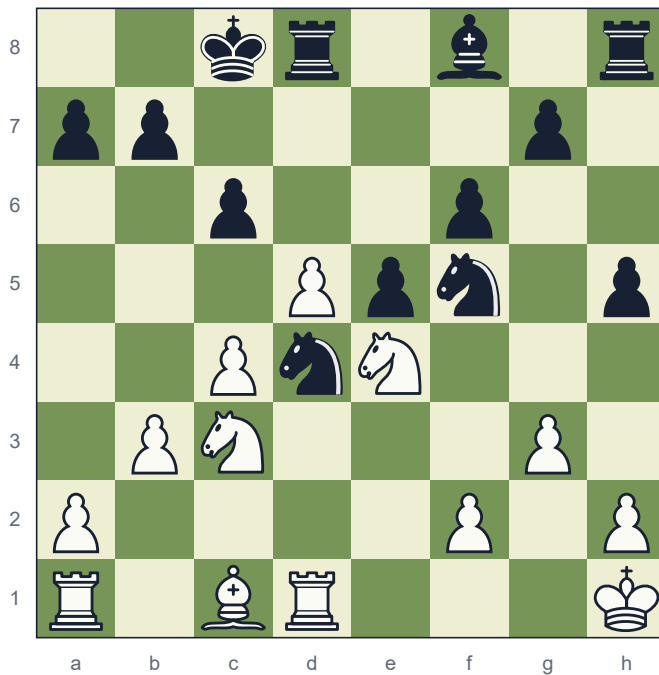
EXPLAIN THE PREFERRED MOVE

Places the knight on d2. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-04 / decision 3



COMPARE CANDIDATES

Rb1 and h4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king h1; central-file pawns f2, c4, d5. Black king c8; central-file pawns e5, c6, f6. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE RB1

19. Rb1 Bb4 Search estimate: -0.28 from White's perspective.

CANDIDATE H4

19. h4 Bb4 20. Kg2 Bxc3 21. Nxc3 cxd5 22. Nxd5 Nc2 Search estimate: -0.26 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Places the rook on b1. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-05 / Choose c4 before the complete shell



Laboratory start - White to move

OPENING BACKGROUND

White challenges d5 with active development and a castled king. This useful departure resembles a Reti or reversed King's Indian structure more than a complete Hippo.

READ THE CURRENT POSITION

White king g1; central-file pawns e2, f2, d3. Black king g8; central-file pawns e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

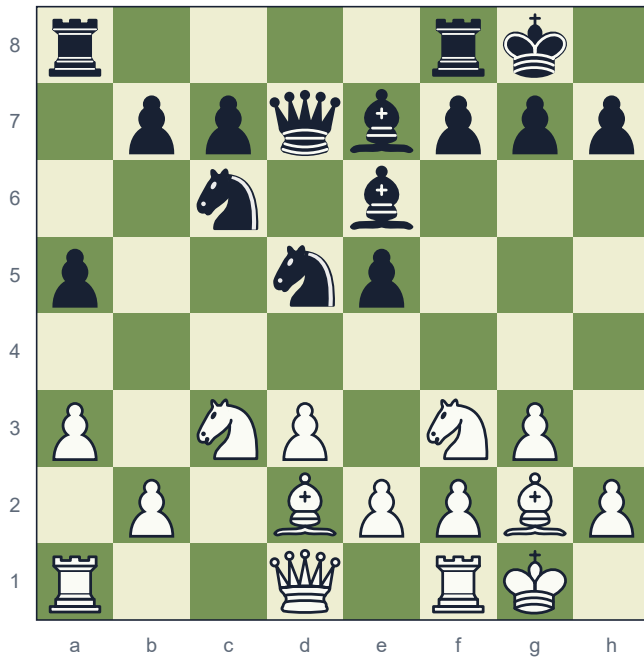
If Black closes with ...d4, Nc3 may be restricted. Plan the knight route before fixing every pawn. Decide which parts still apply after the moves already played.

BEFORE PLAYING

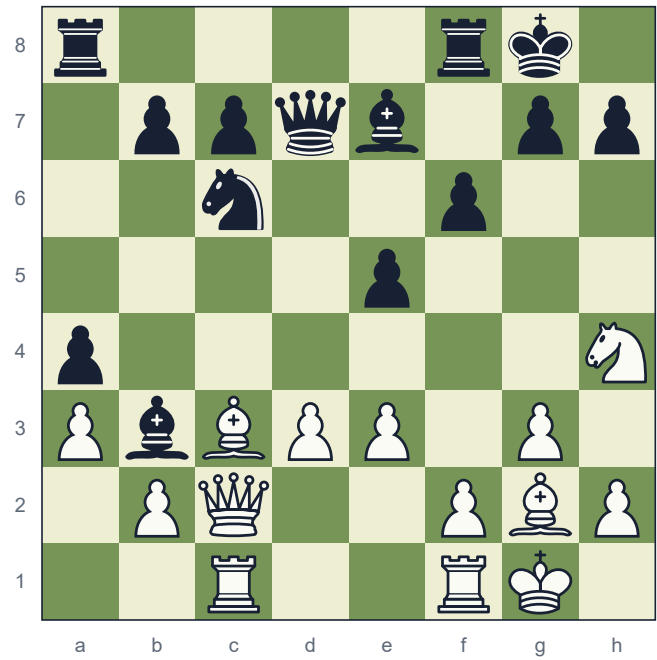
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-05 / play and explain 1

11. Rc1 Nxc3 12. Bxc3 f6 13. Qc2 a4 14. e3 Bd5 15. Nh4 Bb3



Before this segment - White to move



After this segment - White to move

11. RC1

Places the rook on c1. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

13. QC2

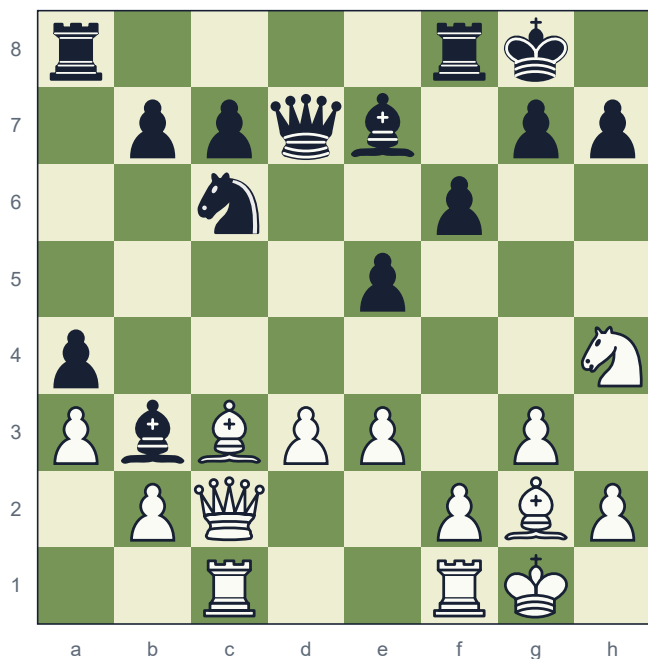
Places the queen on c2. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

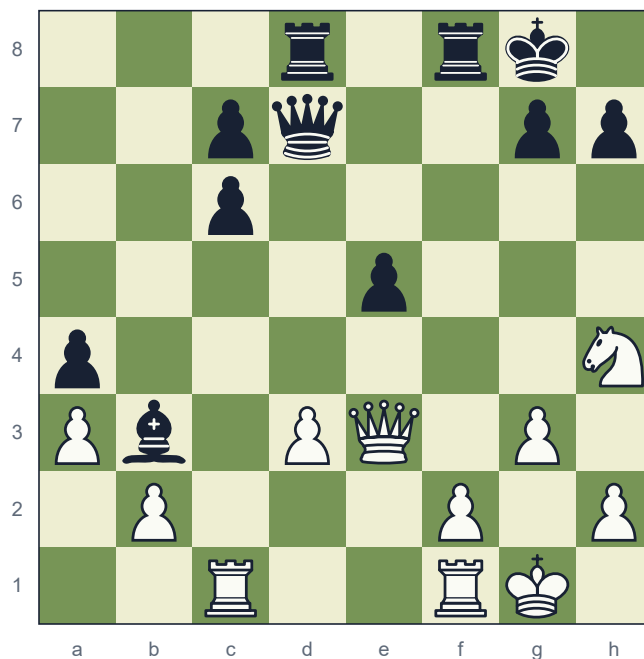
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-05 / play and explain 2

16. Qe2 Rad8 17. Be4 Bc5 18. Bxe5 Bxe3 19. Bxc6 bxc6 20. Qxe3 fxe5



Before this segment - White to move



After this segment - White to move

16. QE2

Places the queen on e2. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

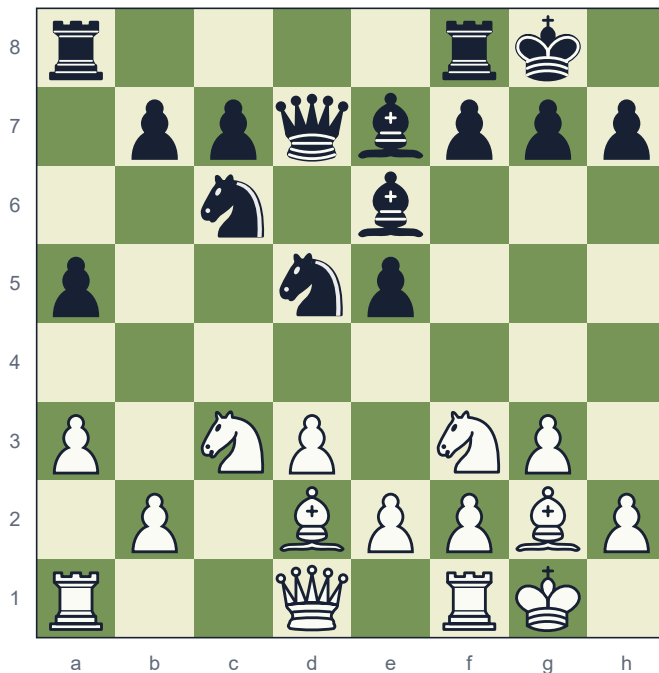
18. BXE5

The bishop captures a pawn on e5. Calculate the best reply before assuming a recapture, then recount material and opened lines.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-05 / decision 1



White to move

COMPARE CANDIDATES

Rc1 and Re1. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns e2, f2, d3. Black king g8; central-file pawns e5, c7, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE RC1

11. Rc1 Search estimate: -0.08 from White's perspective.

CANDIDATE RE1

11. Re1 Nxc3 12. Bxc3 f6 13. Qa4 Rfd8 14. Nd2 Nd4 Search estimate: -0.08 from White's perspective.

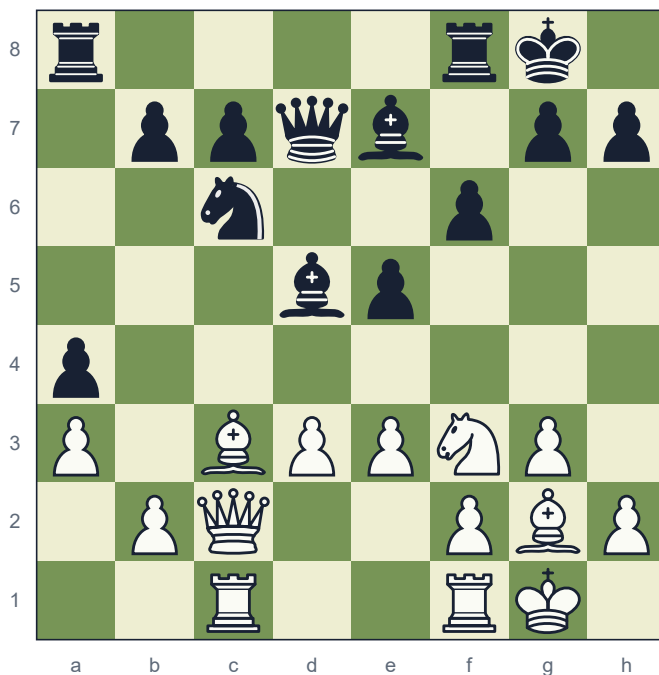
EXPLAIN THE PREFERRED MOVE

Places the rook on c1. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-05 / decision 2



COMPARE CANDIDATES

Nh4 and Rfe1. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, d3, e3. Black king g8; central-file pawns e5, f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NH4

15. Nh4 Bb3 Search estimate: -0.10 from White's perspective.

CANDIDATE RFE1

15. Rfe1 Nd8 16. Nh4 Bxg2 17. Kxg2 g5 18. Nf3 g4 Search estimate: -0.16 from White's perspective.

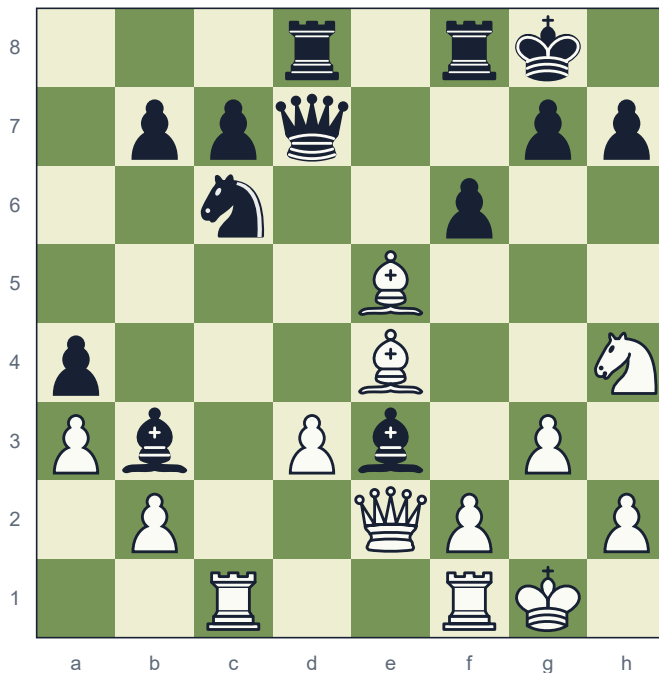
EXPLAIN THE PREFERRED MOVE

Places the knight on h4. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-05 / decision 3



COMPARE CANDIDATES

Bxc6 and fxe3. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, d3. Black king g8; central-file pawns f6, c7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BXC6

19. Bxc6 bxc6 Search estimate: +0.04 from White's perspective.

CANDIDATE FXE3

19. fxe3 Nxe5 20. d4 Nf7 21. Qd3 Ng5 22. Bxb7 c5 Search estimate: -0.24 from White's perspective.

EXPLAIN THE PREFERRED MOVE

The bishop captures a knight on c6. Calculate the best reply before assuming a recapture, then recount material and opened lines.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-06 / Choose e4 for a KIA transition



Laboratory start - White to move

OPENING BACKGROUND

White adopts a King's Indian Attack center because Black's structure makes it useful. The b-pawn remains flexible instead of being committed automatically.

READ THE CURRENT POSITION

White king g1; central-file pawns f2, c3, d3, e4. Black king g8; central-file pawns c5, d5, e6, f7. Compare the pawn contacts and opened files before choosing a move.

OPENING CONCERN TO RECHECK

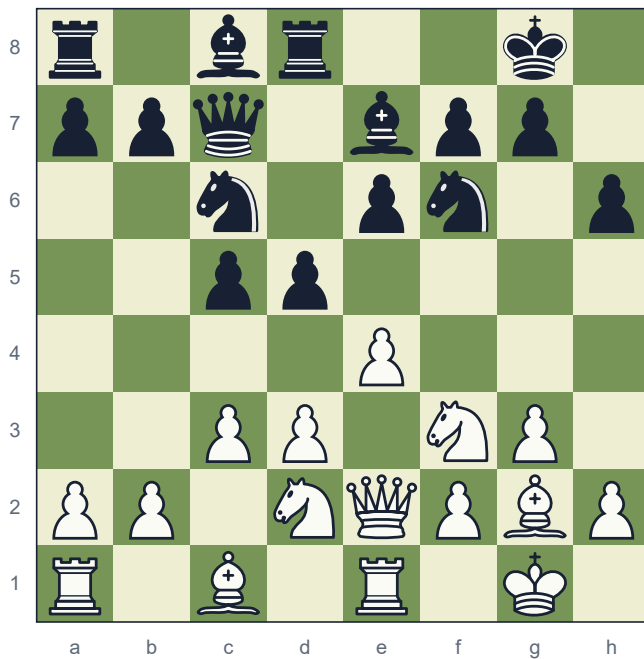
Black can exchange on e4 or advance ...d4. Do not assume the e5 attack is available in every branch. Decide which parts still apply after the moves already played.

BEFORE PLAYING

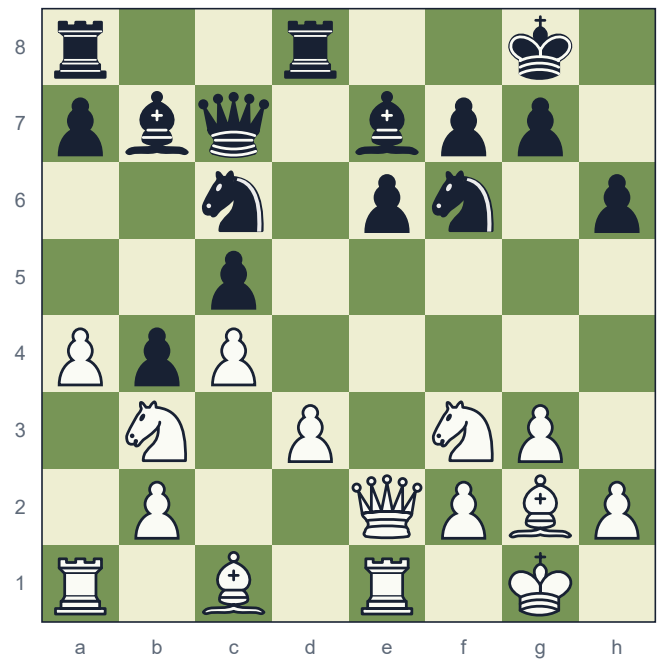
Write a candidate move and a reply for the opponent. Name the pawn break each side would like to achieve. The continuation on the following pages is one checked illustration, not a forced script.

HW-LAB-06 / play and explain 1

11. a3 b5 12. exd5 Nxd5 13. a4 b4 14. c4 Nf6 15. Nb3 Bb7



Before this segment - White to move



After this segment - White to move

11. A3

Controls nearby wing squares and may restrain a piece or pawn. Compare this useful purpose with the time lost to central development.

13. A4

Places the pawn on a4. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

REASSESS

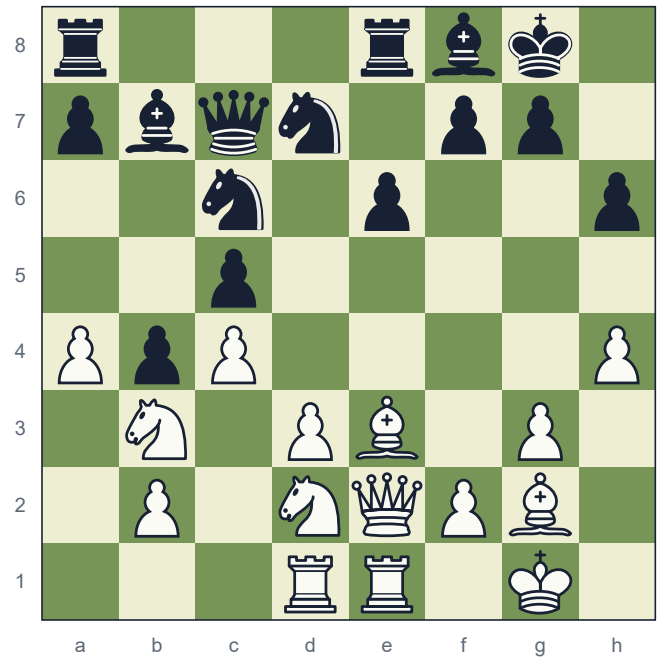
Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-06 / play and explain 2

16. Bf4 Qc8 17. Rad1 Re8 18. h4 Bf8 19. Be3 Nd7 20. Nfd2 Qc7



Before this segment - White to move



After this segment - White to move

16. Bf4

Places the bishop on f4, attacking queen on c7. A legal capture may still be unsound if the target is defended.

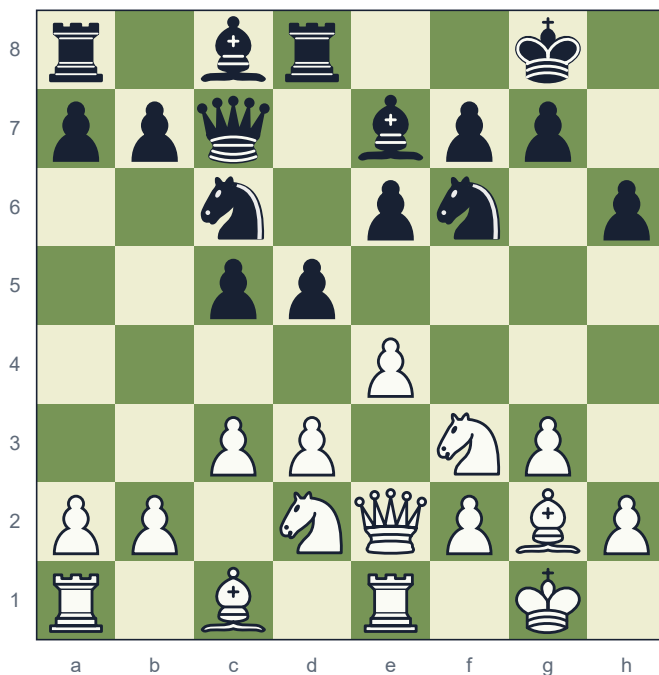
18. H4

Creates a wing or central pawn contact and changes king shelter. Calculate captures and the opponent's central counterplay before extending the advance.

REASSESS

Name the material balance, the most active piece and the next pawn break. If the center changed, explain why the original plan may need to change too.

HW-LAB-06 / decision 1



COMPARE CANDIDATES

a3 and a4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, c3, d3, e4. Black king g8; central-file pawns c5, d5, e6, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE A3

11. a3 b5 12. exd5 exd5 13. d4 c4 14. Ne5 Re8 Search estimate: +0.10 from White's perspective.

CANDIDATE A4

11. a4 Rb8 Search estimate: +0.03 from White's perspective.

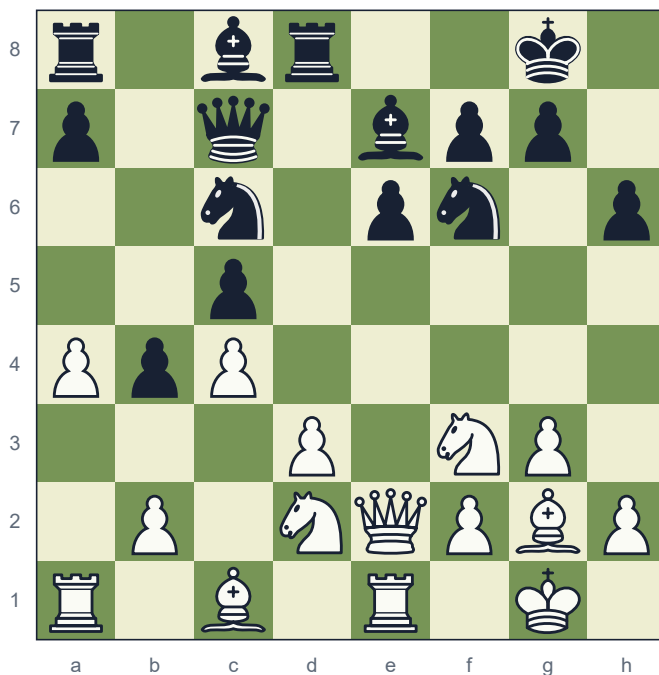
EXPLAIN THE PREFERRED MOVE

Controls nearby wing squares and may restrain a piece or pawn. Compare this useful purpose with the time lost to central development.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-06 / decision 2



White to move

COMPARE CANDIDATES

Nb3 and a5. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, d3, c4. Black king g8; central-file pawns c5, e6, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE NB3

15. Nb3 Bb7 16. Bf4 Qc8 17. a5 a6 18. Rad1 Bf8 Search estimate: +0.18 from White's perspective.

CANDIDATE A5

15. a5 Nxa5 Search estimate: +0.04 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Places the knight on b3. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.

HW-LAB-06 / decision 3



White to move

COMPARE CANDIDATES

Be3 and d4. Before reading the continuations, explain a useful purpose for each and calculate the opponent's most forcing reply.

CURRENT POSITION

White king g1; central-file pawns f2, d3, c4. Black king g8; central-file pawns c5, e6, f7. Compare the pawn contacts and opened files before choosing a move.

CANDIDATE BE3

19. Be3 Nd7 20. Nfd2 f5 21. Nf3 a5 22. h5 Nd8 Search estimate: +0.06 from White's perspective.

CANDIDATE D4

19. d4 cxd4 Search estimate: -0.04 from White's perspective.

EXPLAIN THE PREFERRED MOVE

Places the bishop on e3. Compare its activity and safety with its previous square, then use the branch explanation to connect the move to a plan.

HOW TO INTERPRET THIS

The scores compare these candidates in one finite search. Close values mean the explanation and practical comfort matter; they do not imply a unique winning move. Replay both lines and identify the first concrete difference.